

ALG-02 - Teacher Diagnostic Key

Recognition lab

1. $(x-4)^2+4$.
2. $12-(x+1)^2 \leq 12$.
3. AM-GM on positive $x, 25/x$; equality $x=5$.
4. $(a-b)^2 \geq 0$ or AM-GM $\rightarrow ab \leq 81$, equality $a=b=9$.
5. No; 0 is excluded and values approach it.
6. Real equality point is $1/2$; integer filter checks 0,1.
7. Engel: $1/x+1/y \geq 4/(x+y)=1/3$.
8. Equality is excluded, so 7 is only a lower bound unless another point attains it; no minimum 7.
9. No. Root-count/feasibility by discriminant belongs to ALG-03.
10. Complete the square.
11. AM-GM: $a+b \geq 6$, equality $a=b=3$.
12. No positive minimum; product approaches 0 as one variable approaches $0+$.
13. Equality condition and attainment in the original domain.
14. Check adjacent admissible integers (normally floor/ceiling candidates, plus any boundary constraints).
15. Upper bound 1 is approached as $x \rightarrow \infty$ but never attained; no maximum.
16. No extremum claim is needed; only proof and equality characterization if requested/useful.

Practice answers

1. $(x-3)^2+4$, min 4 at $x=3$.
2. max 9 at $x=4$.
3. $x+4/x \geq 4$, equality $x=2$.
4. $ab \leq 25$, equality $a=b=5$.
5. On $x>0$, lower bound 0 is not attained; no minimum.
6. min 6 at $x=3$.
7. max 36 at $a=b=6$.
8. min $2/5$ at $x=y=5$.
9. $(x+5)^2+4$, min 4.
10. $9/4$, attained at $n=0,1$.
11. min 8 at $x=y=4$.
12. max 100 at $x=y=10$.
13. no minimum; infimum 0.
14. no maximum; supremum 1.
15. $n^2-5n+9=(n-5/2)^2+11/4$; integers 2,3 give 3.
16. perimeter 40 $\rightarrow x+y=20$; max area 100 at 10,10.
17. Let $u=2p, v=8q$; $uv=16pq=576$, so $u+v \geq 48$; min 48 when $2p=8q$, giving $p=12, q=3$.
18. Engel $\rightarrow 1/x+1/y \geq 4$, equality $x=y=1/2$.
19. exact: $t^2+t^2-2=(t+t^2-1)^2-2=23$; generic AM-GM lower bound is 2 and is not the requested exact consequence.
20. balance integers 7,8; max 56.
21. Put $s=x+y$. $(x-y)^2=2(s+1012)-s^2 \geq 0$; positive root bound is <47 , so largest integer 46.
22. AM-GM $a+b+c \geq 3(abc)^{1/3}=3$; equality $a=b=c=1$.
23. $x^2/y+y^2/x = (x^3+y^3)/(xy)$; with $x+y=8$, symmetric reduction or Engel gives minimum 8 at $x=y=4$. Direct Engel: $x^2/y+y^2/x \geq (x+y)^2/(x+y)=8$.
24. real form $(n-7/2)^2+11/4$; integer min 3 at $n=3,4$; real min $11/4$.
25. $x+1/x > 2$ for $x>1$, and approaches 2 as $x \rightarrow 1+$; no minimum, infimum 2.
26. Square completion answers value/location of vertex; discriminant canon answers root behavior/feasibility.
27. real: min 0 at 0; positive real: no minimum, infimum 0; nonzero integer: min 1 at $+1$.
28. Inequality created a finite feasible interval; discrete/divisibility filters still decide admissible integers.
29. area xy under fixed $x+y$ is maximized at equality; geometry supplies the context.
30. Any correct example, e.g. x^2 on $x>0$.

H0 answers

1. $(x-6)^2+4$; min 4.
2. max 17 at $x=5$.
3. min 12 at $x=6$.
4. max 64 at $a=b=8$.
5. min $2/7$ at $x=y=7$.
6. no minimum; infimum 0.
7. minimum $5/4$ at $n=1,2$.
8. $(n-9/2)^2+11/4$; integer min 3 at $n=4,5$.
9. 10 is a lower bound but not attained under the stated equality mechanism; cannot claim minimum 10.
10. max 81 at $p=q=9$.
11. no minimum; infimum 0.
12. ALG-03; repeated-root/discriminant behavior is polynomial-root canon.
13. Evaluate the objective at admissible integers nearest 2.4 (and relevant boundaries), compare, then verify constraints.
14. $x+y=14$; max area 49 at 7,7.
15. The inequality is a lower bound on the sum, not an upper bound. With fixed product, the sum is unbounded above.
16. Example $f(x)=1-1/x$ on $x>0$: supremum 1, no maximum.

Diagnostic tags

- ALG02-R1 bound called extremum without equality.

- ALG02-R2 equality condition identified but not checked against domain.
- ALG02-R3 inequality direction does not match request.
- ALG02-R4 continuous optimum accepted in discrete domain.
- ALG02-R5 named theorem selected without matching structure/hypotheses.
- ALG02-R6 discriminant/root doctrine imported from ALG-03.